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NPTEL ONLINE CERTIFICATION COURSE

SUSTAINABLE TRANSPORTATION SYSTEMS

Lecture 32: Sustainable Transportation Planning & Approaches-III
Management Strategies for Transportation Systems

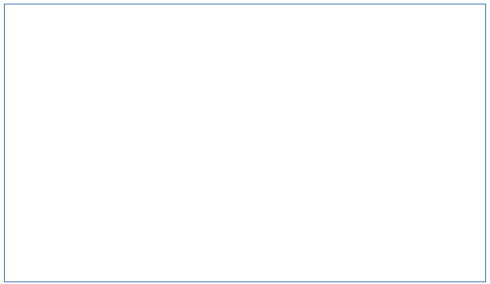
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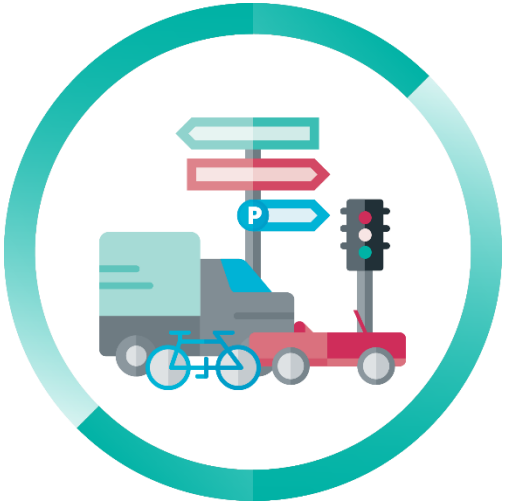


Transportation System Management

- Solution to Peak hour congestion, delay and accidents in cities

- Widening of roads, providing elevated flyovers or constructing bypasses and urban expressways may prove costly.

- Transportation System Management (TSM) is a set of measures, which are adopted to ensure the most productive and cost effective use the existing transportation services and modes



Source: (L. R. Kadiyali, 2017)

Urban Traffic Management Strategies

Turning Restrictions on Streets for Vehicles







Prohibited Right/Left or 'U' turning movements

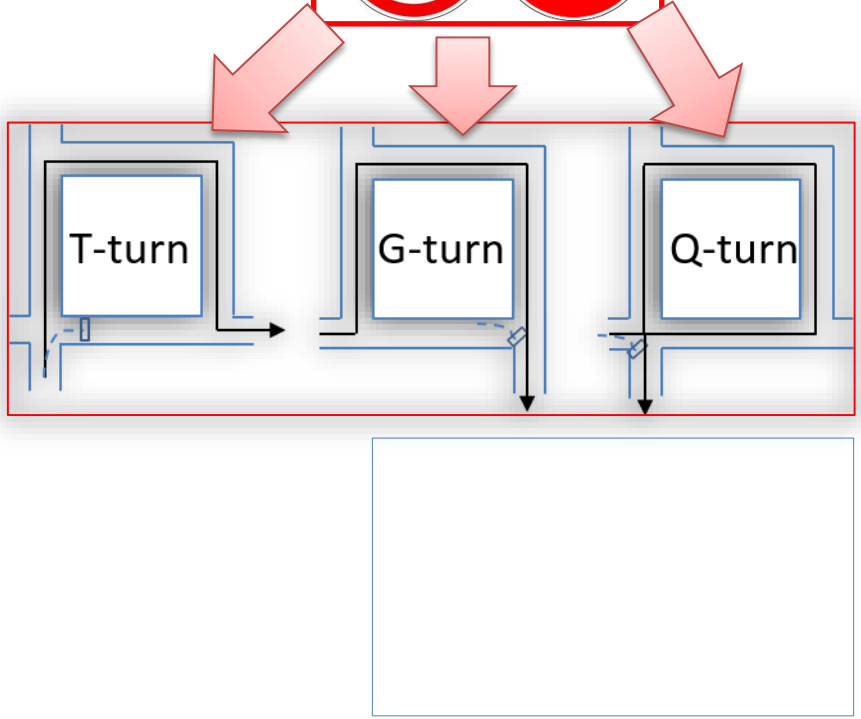
- Can be established only if the street is capable of alternate routing.
- Known to increase saturation flow and capacity at an intersection.
- This method cannot solve congestion whole together, but can assist other techniques to reduce congestion efficiently.

Source: (L. R. Kadiyali, 2017)

Prohibited Right/Left turning movements



- The Left/Right turning movements are replaced by either a T-turn, G-turn or Q-turn
  - Ex. T-turn: Diversion of right-turning traffic to an alternative intersection capable of dealing with more traffic.
  - G-turn: Diversion of right-turning traffic to the left before the junction.
  - Q-turn: Diversion of right-turning traffic beyond the junction.



Source: (L. R. Kadiyali, 2017)

Example of Turn Restrictions in the City of Menlo Park



An example of Turn Restrictions in the Willows neighborhood of the City of Menlo Park, California

Source: (https://menlopark.org/DocumentCenter/View/25903/J1-20200811-CC-Willow-turn-restrictions)

Overall Traffic Volume reduction in Menlo Park City

	Northbound or Eastbound			Southbound or Westbound			Total		
	2017	2019	% ▲	2017	2019	% ▲	2017	2019	% ▲
Durham Street (at Willow Road)	236	44	-81%	64	34	-47%	300	78	-74%
Right-turn from Durham Street to Willow Road (prohibited in 2019 data)	218	17	-92%	-	-	-	218	17	-92%
Willow Road (at Durham Street)	470	1,042	+122%	694	569	-18%	1,164	1,611	+38%
Right-turn from Willow Road to Durham Street	5	4	-20%	-	-	-	5	4	-20%

Source: City of Menlo Park

Source: (https://menlopark.org/DocumentCenter/View/25903/J1-20200811-CC-Willow-turn-restrictions)

One-way Streets

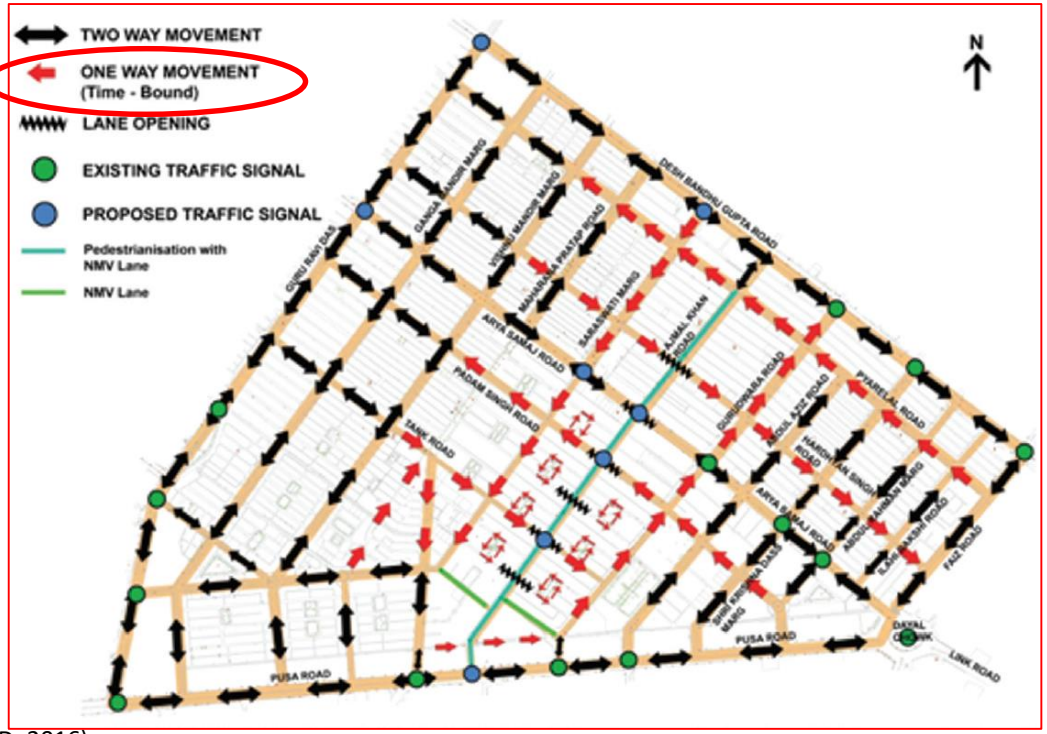
- Streets with movements permitted only in one direction.
- Intended to improve traffic flow, increase capacity and reduce delays
- Least expensive strategy and yields beneficial results when combined with banned turning movements, installation of signals etc.





Source: (L. R. Kadiyali, 2017)

Example of Implementation of One-way Circulation system Plan of Karol Bagh area, New Delhi







Source: (MOUD, 2016)

Car pooling

- Allows travelers to share ride from/to a common origin/destination.
- May be
  - Casual or informal car pooling
  - Real-time Car pooling: App-based
  - Van pooling: 7 to 15 passengers sharing the cost of a van as a daily mode.



- Reduces private vehicles on streets, reduces congestion on streets, fuel economic benefits to passengers

Source: (Shaheen et. al, 2018)

Car pooling: Social benefits

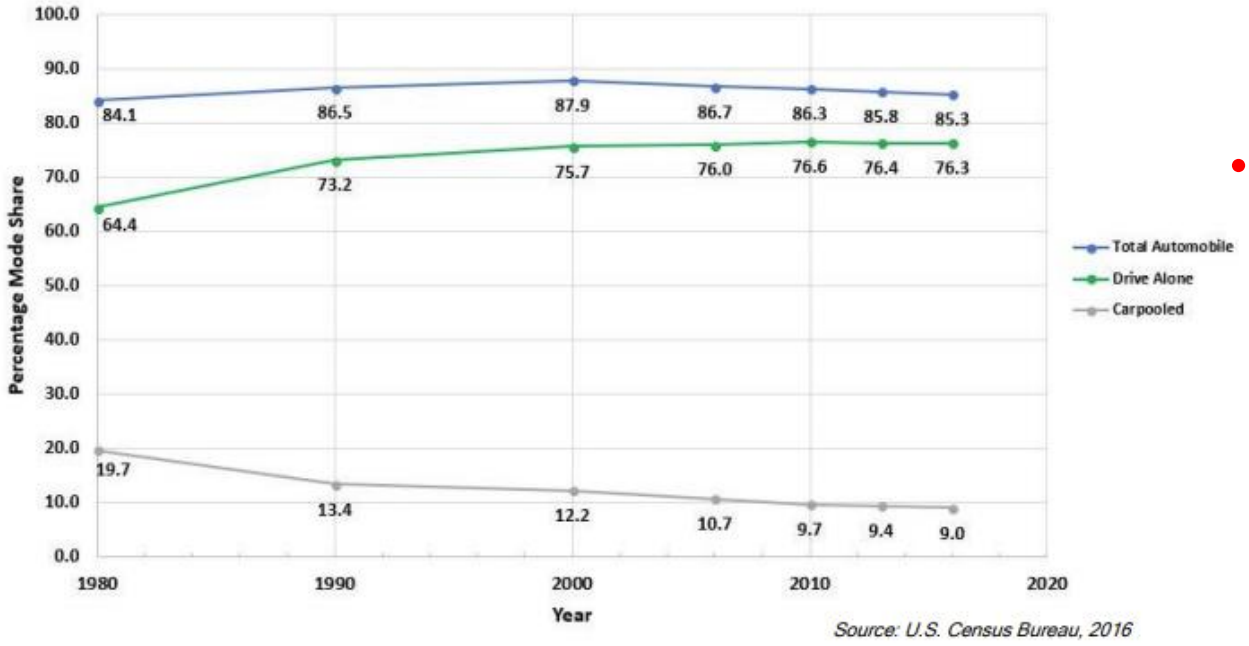




- Reduced vehicle miles travelled
- Reductions in fuel consumption and greenhouse gas (GHG) emissions
- Reductions in adverse air pollution impacts
- Cost savings for public agencies and employers

Source: (Shaheen et. al, 2018)

Example of Car pooling in the U.S



- Car pooling is the 2nd most common travel mode to work in the U.S after driving alone
- Decrease in mode share due to car pooling in the U.S.

Source: (Shaheen et. al, 2018)

Example of Promoting Car Pooling in the U.S



- A Morning rush hour in the Dale City.
- Left Lanes: Open to all Traffic
- Right Lanes: Open only to buses and cars with minimum three riders

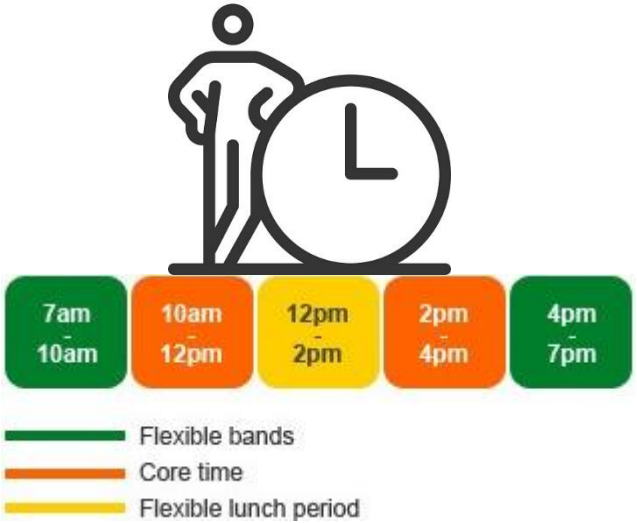
Source: (https://www.nytimes.com/2011/01/29/us/29carpool.html)

Staggering of Office & School hours

- Flexibility for the employees to choose their office hours keeping the core time fixed.
  - Ex. Working time from 10 A.M – 7 P.M or from 7 A.M – 4 P.M

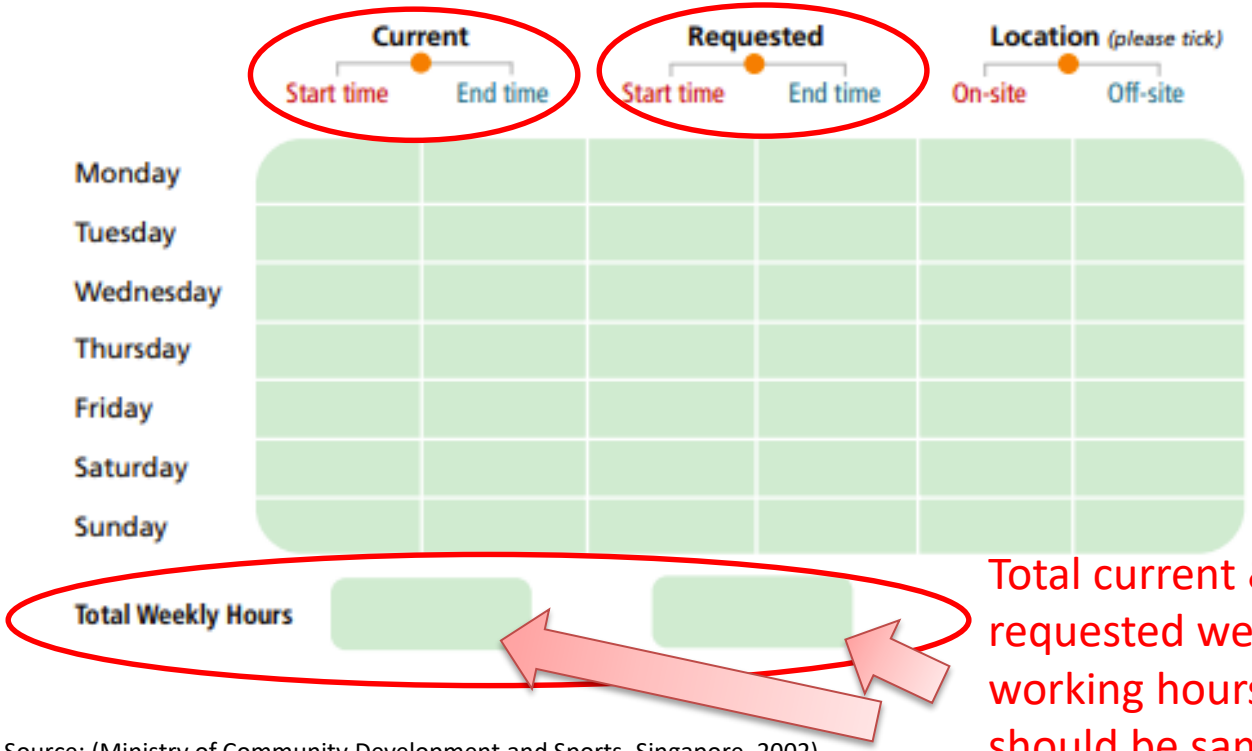
- Can reduce the peak-hour traffic on streets to a great extent by reducing congestion and improving public transport conditions on streets

- Same strategy can be adopted for School hours also.



Source: (Genevieve Giuliano and Thomas F. Golob, 1990)

Sample of a Staggered Office Working time



Source: (Ministry of Community Development and Sports, Singapore, 2002)

Total current & requested weekly working hours should be same

- The main idea is the flexible morning and evening hours keeping the core time and the Total weekly hours same.

Cost Benefit Savings due to Staggered School hours in Linkoping Town, Sweden

	Later school day		Staggered school day	
Effect	**Benefit**	**Cost**	**Benefit**	**Cost**
Buses not needed	8		8	
Costs for the pupils		8.5		11.3
Benefit for pupils who want the change	x		x	
Costs for the teachers		x		x
Costs for the principal (harder to schedule classes)				x
Sum	**8**	**8.5**	**8**	**11.3**

- Cost-Benefit analysis of (a) one hour later school day and (b) half an hour staggered start of the school day in Million SEK per year
- x denotes the non-valued effects

1/3 of the classes starts half an hour earlier
1/3 half an hour later
1/3 starts at the same time as today

1 SEK (Swedish Krona) = 8.61 INR (Indian Rupee)

Source: (Ministry of Community Development and Sports, Singapore, 2002)

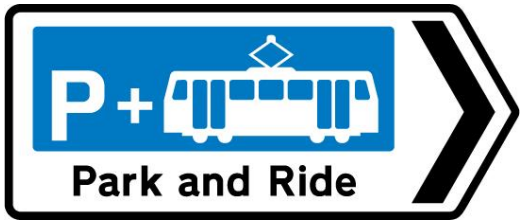
Peripheral Parking schemes

- Park and ride facilities provides the option for public to park their private vehicle at city outskirts and choose public mode of transport to travel to city centre.

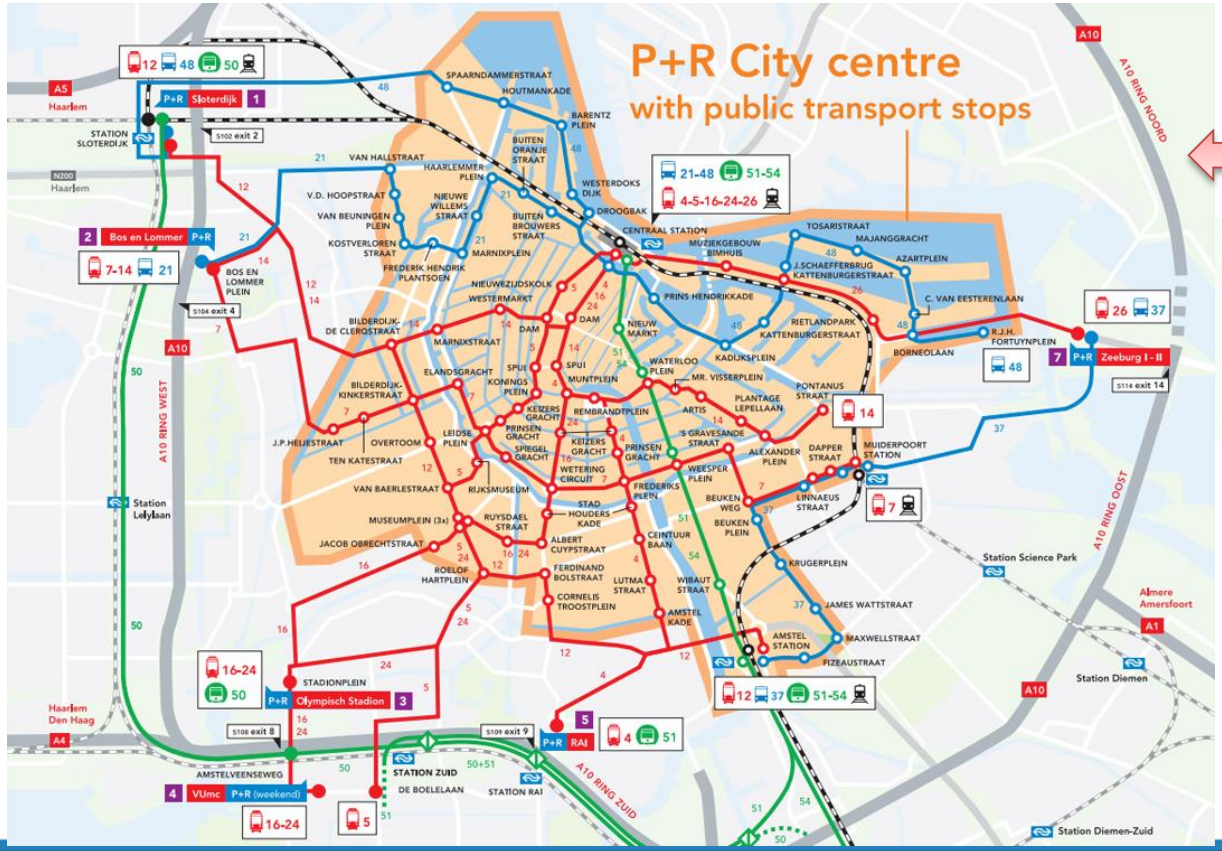
- Should be integrated with good public transport facilities such as car pooling, BRTS, metro etc.

- Reduces fuel cost for passengers, free from tiredness due to congestion in city centres





Example of Park and Ride facilities in Amsterdam City



Source: (Maps Amsterdam)

Chartered buses (Institutional buses)

- Chartered bus services or institutional buses connecting the urban centres
- Trips made from city centre to Business parks, schools and colleges, government offices etc.
- Reduce additional private vehicle inflow to streets, reduced congestion, reduced fuel usage.
- Good connectivity to city centre should be given.



Internal Shuttle in the CBD

- Uninterrupted shuttle bus services plying on all routes within the city
- To provide better connectivity across the city.
- Integration with the Park and Ride facilities could increase demand of this mode.
- Integration with mobile applications and Real time Information will increase the interest for passengers.



Source: (L. R. Kadiyali, 2017)

Example of the National University of Singapore (NUS): Institutional & Shuttle Bus Services

Services	(Mondays to Saturdays)	(Sundays & Public Holidays)	Frequency
A1	First Bus - 7:15am Last Bus – 11.00pm	First Bus – 9.00am Last Bus – 11.00pm	7.15am to 7.00pm – 15mins 7.00pm to 11.00pm – 30mins
A2	First Bus - 7:15am Last Bus – 11.00pm	First Bus – 9.00am Last Bus – 11.00pm	7.15am to 7.00pm – 15mins 7.00pm to 11.00pm – 30mins
B1	No Service	No Service	No Service
B2	No Service	No Service	No Service
C	First Bus - 7:30am Last Bus – 7.00pm	No Service	7.30am – 7.00pm – 30mins
D1	First Bus - 7:15am Last Bus – 11.00pm	First Bus – 9.00am Last Bus – 11.00pm	7.15am to 7.00pm – 15mins 7.00pm to 11.00pm – 30mins
D2	First Bus - 7:15am Last Bus – 11.00pm	First Bus – 9.00am Last Bus – 11.00pm	7.15am to 7.00pm – 15mins 7.00pm to 11.00pm – 30mins

Service	(Mondays to Fridays)	(Saturdays, Sundays & Public Holidays)	Frequency
BTC	First Bus - 7:30am Last Bus – 9.30pm	No Service	7.30am to 9.30pm – 60 mins



Source: (National University of Singapore)

Congestion pricing

- Demand-based strategy
- To encourage private vehicles to shift away from congested routes during peak hours.
- Private vehicles using the route during peak hours are fined heavily.



Source: (PIARC Technical Committee on Road Network Operations)

Advantages of Congestion pricing

- Reduce congestion during peak hours
- Efficient use of transportation system, more people use alternate routes or shift to public modes
- Smooth traffic flows saving time and delay
- Revenue generation
- Reduced accidents on roads
- Reduced emissions from vehicles and idling



Source: (PIARC Technical Committee on Road Network Operations)

Pricing example 1: Cordon pricing in London



Source: (Jean-Paul Rodrigue 2020)

London have decided to charge its private vehicles for entry into the Central London in early 2003.

- Proved to be successful despite a great deal of opposition

Source: (https://ops.fhwa.dot.gov/publications/fhwahop08047/02summ.htm)

Pricing example 2: Singapore





Singapore CBD Pricing Zone

Source: (https://ops.fhwa.dot.gov/publications/fhwahop08047/02summ.htm)

Source: (Jean-Paul Rodrigue 2020)

Singapore have implemented extreme measures such as limiting car purchases, high licence vehicles, electronic tolls on highways and cordon pricing in the downtown area.

Parking restrictions

- Parking restrictions within the city centre indirectly reduces vehicle flow into the city

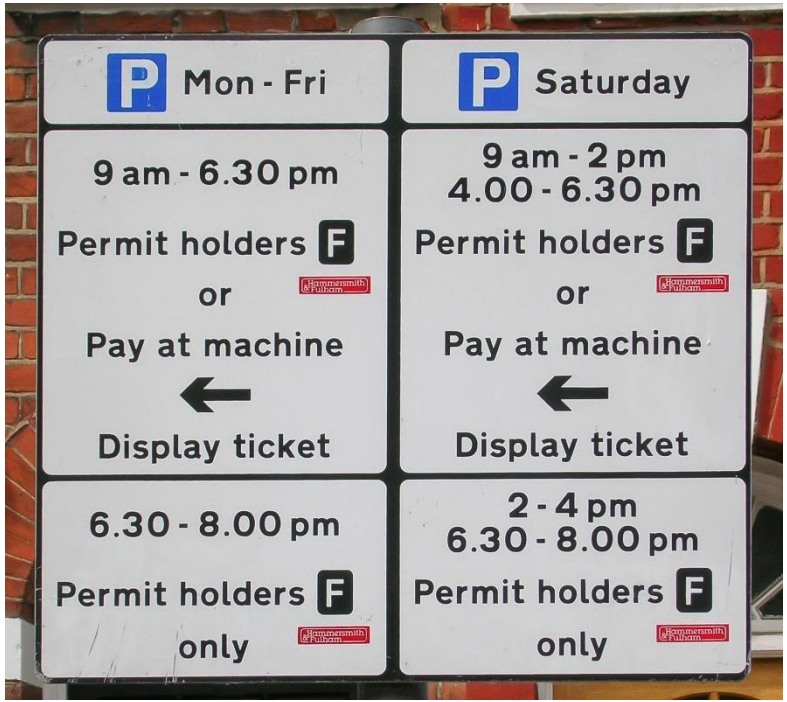
- Benefits includes
  - Increased automobile occupancy levels on streets
  - Decrease in person trips
  - Faster travel times and decreased delays
  - Increase in public transport use
  - Reduction in Air pollution
  - Low ambient noise levels



Source: (L. R. Kadiyali, 2017)

Example of London as a Controlled Parking Zone (CPZ)

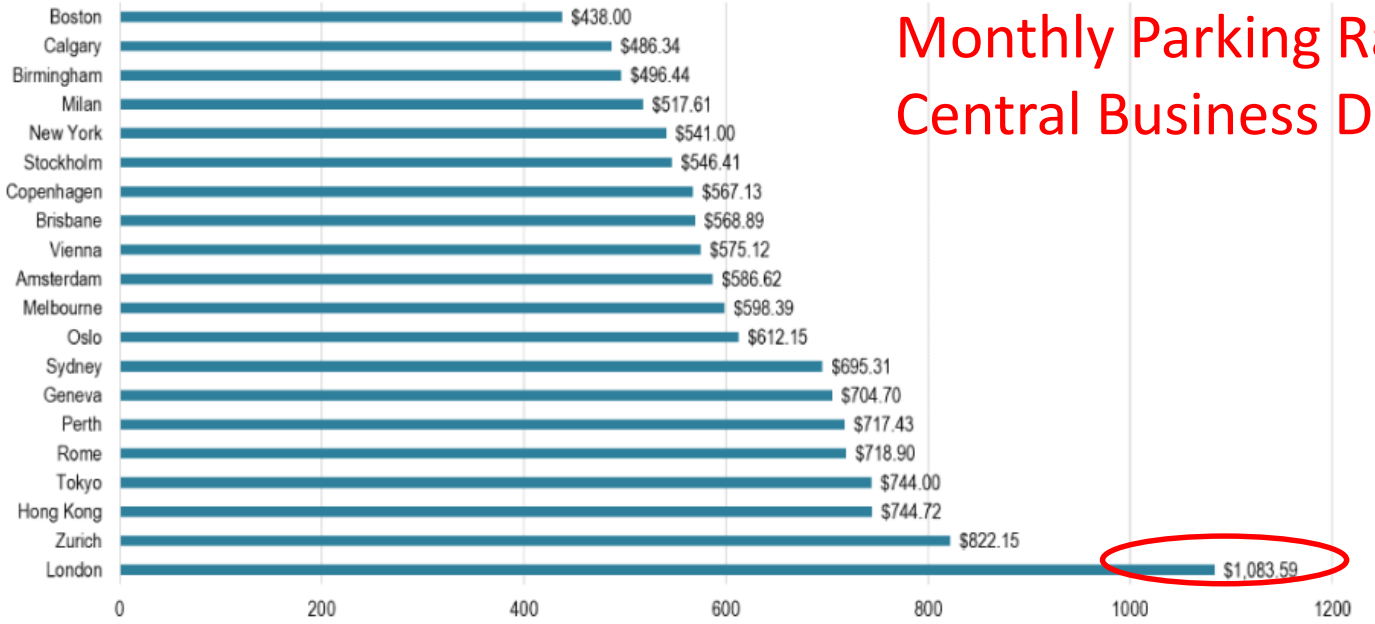
The whole of the City of London is a Controlled Parking Zone (CPZ)



- In a CPZ all parking is controlled. Parking is controlled by hours and allowed only in designated parking bays.
- The controlled parking hours in a CPZ are
  - Monday to Friday **7am - 7pm**
  - Saturday **7am - 11am**

Source: (https://www.cityoflondon.gov.uk/services/parking/parking-restrictions)

Parking Pricing in CBD Worldwide



Monthly Parking Rates in the Central Business Districts, 2011

Source: (Jean-Paul Rodrigue 2020)

Transport Rationing for transport management technique

- Traffic rationing is part of transport management.
- Rationing restricts the no. of vehicles or its movement through various measures.
- No drive day, driving restriction, odd-even schemes are example of transport rationing.

Odd-Even scheme in Delhi

- The Odd-Even Policy is a transport rationing mechanism which is a part of transport management.
- Theoretically it control pollution levels through reduce traffic & congestion by restricting the number of on-road vehicles per day.
- It was applied in Delhi at times, when ambient air quality deteriorated much.



Image: Volunteer at awareness drive for odd-even scheme.
Source: indiatvnews.com

Odd-Even scheme: Limitations

- Paris, Rome, Bogota, Mexico city and Beijing are some example where this scheme was followed.
- Most of the places, this failed after initial temporarily success.
- Mostly people buy one cheap secondary car for such conditions which defeats whole purpose of scheme.
- In case of low capacity of public transport or poor infrastructure for other modes/means leads to distress among users.

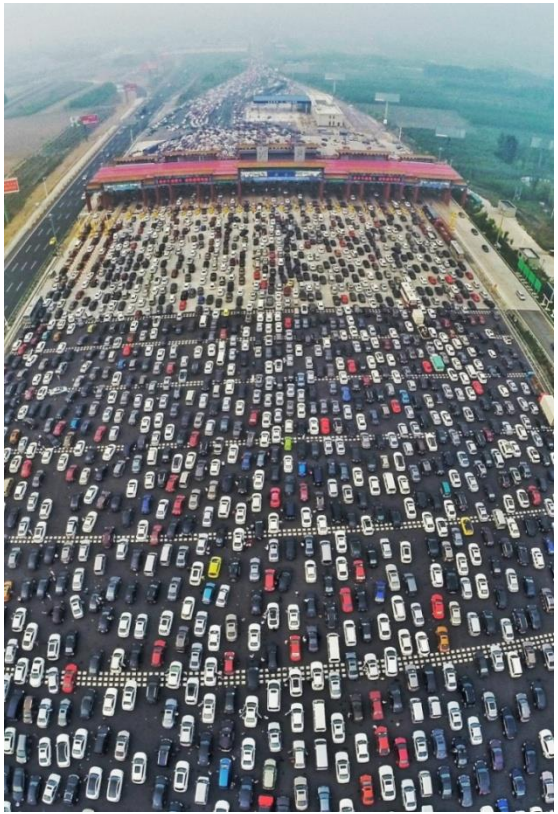


Image: Congested road in Beijing

Source: indiatimes.com

‘Only pedestrian movement’ on weekends

- This is known concept generally applicable in places with very high footfall as market area or recreational area or any tourist place.
- Connaught place is one such area with more then million-foot fall daily.
- NDMC (New Delhi Municipal Corporation) planned to restrict vehicular movement on Saturdays & Sundays in inner circle but met with protest by trader organization.
- Later it rolled back by NDMC.



Image source: mint.com

‘No-Car Sundays’ in Jakarta

- Jakarta is capital city of Indonesia.
- Most days central Jakarta is noisy, polluted and congested with cars and motorcycles.
- But two Sundays a month, people-powered vehicles own the road.
- Program called Car-Free Sunday is growing in popularity in Indonesia's sprawling capital
- In many countries, this concept is gaining traction through civil society including India & Singapore.

Restrictions on entry of trucks during day-time

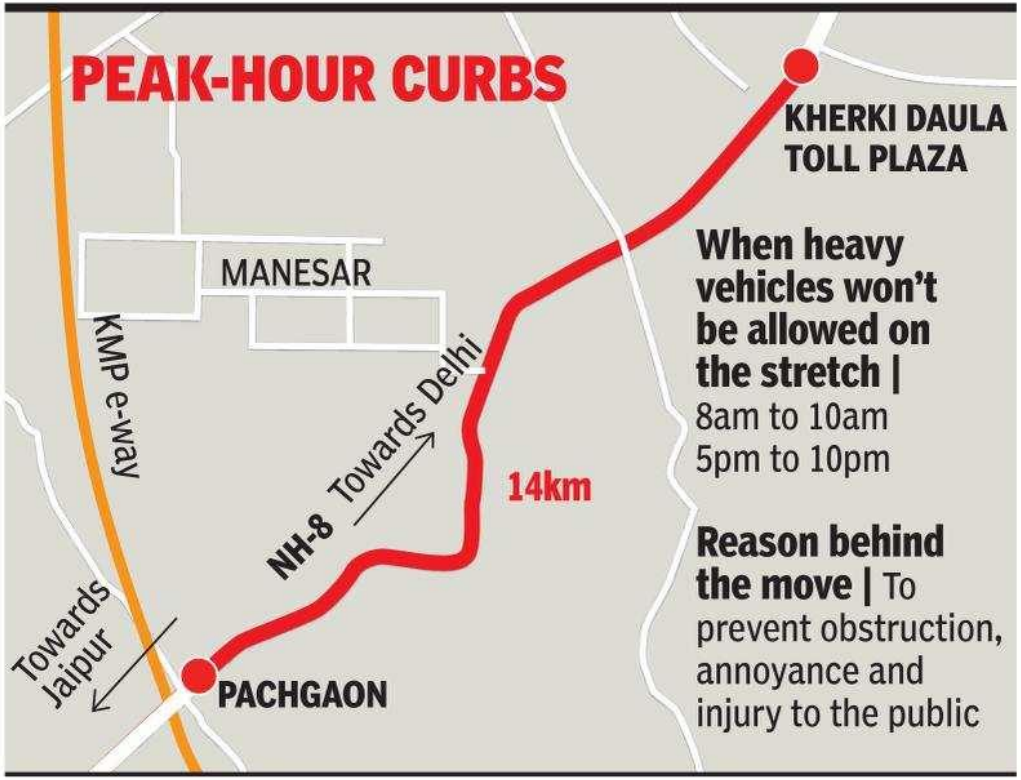
- Time restrictions, is a type of restriction strategy for heavy vehicles on certain roads during peak hours

- Separating heavy vehicles from other modes can achieve an efficient road transport network in terms of:
  - delay time
  - enhanced safety
  - Reduced congestion
  - Reduced noise pollution
  - Reduced air pollution



Source: (Mohammed Al Eisaeia et. al, 2017)

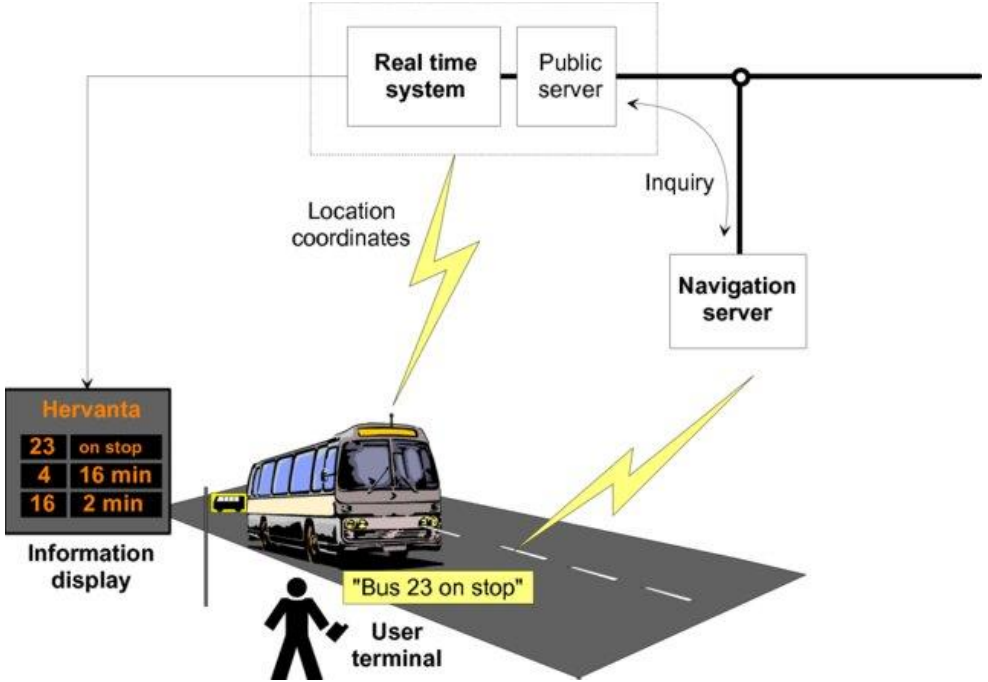
Example of Delhi-Gurugram Expressway

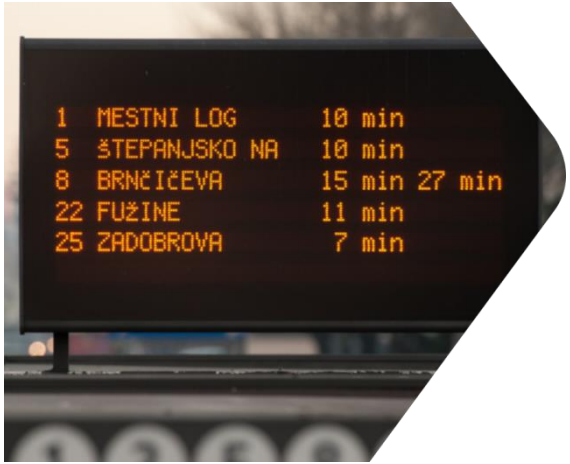


The 14 km stretch of Delhi-Gurugram expressway has no entry for trucks during rush hours (8 A.M to 10 A.M and 5 P.M to 10 P.M)

Source: (The Times of India, 2019)

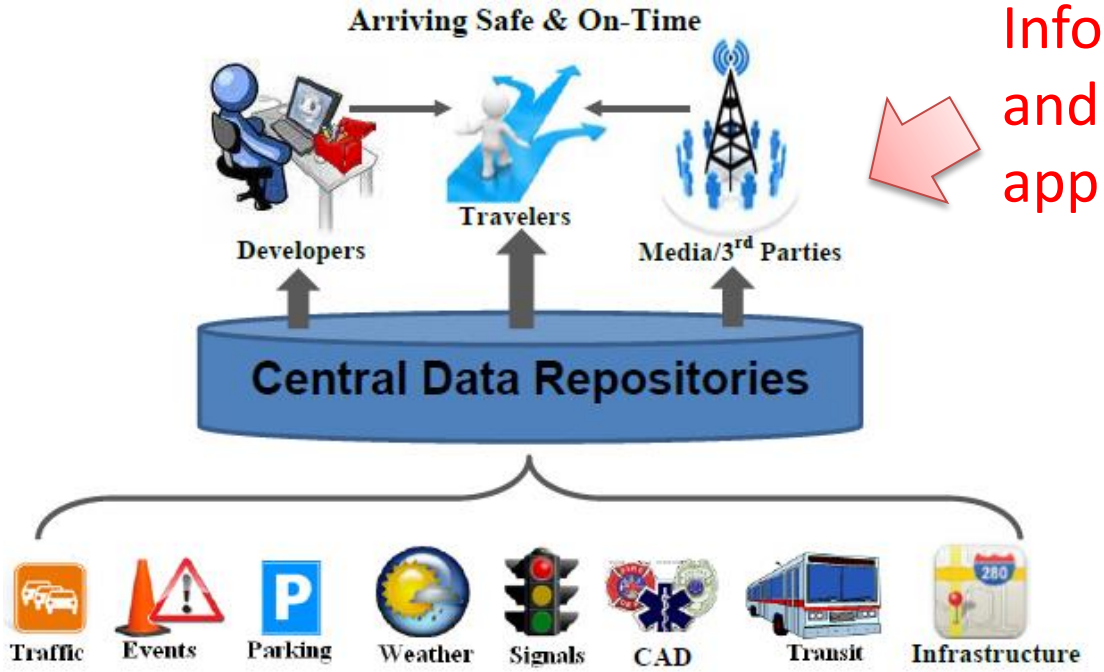
Real-time Travel Information systems





Source: (Koskinen Sami and Viratanen Ari, 2004)

Infrastructure of Real-time Travel Information systems



Information Infrastructure and real-time traveler applications

Source: (NCHRP, 2013)

Benefits of Real-time Travel Information systems

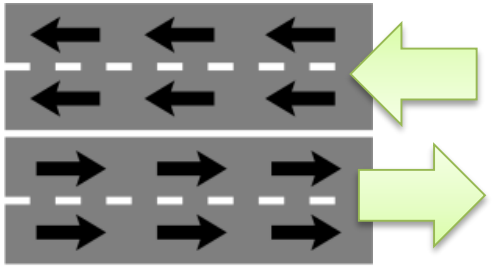
- Increase Transportation system efficiency and capacity
- Enhance mobility
- Improve safety
- Reduce energy consumption and environmental costs
- Increase economic productivity



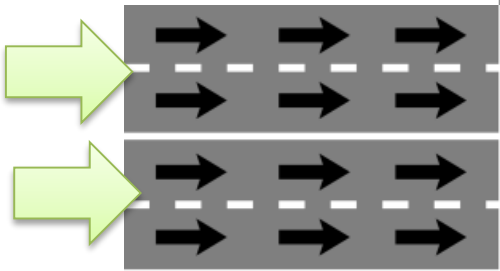
Source: (NCHRP, 2013)

Contraflow lanes

- A lane of traffic made exclusively for peak hour traffic in one direction
- Provides better connectivity, reduces congestion and travel time during peak hours.
- Ex. Single lane of a two way lane towards the business park can be made a contraflow lane in the morning peak hours and lane towards the opposite direction in the evening peak hours.



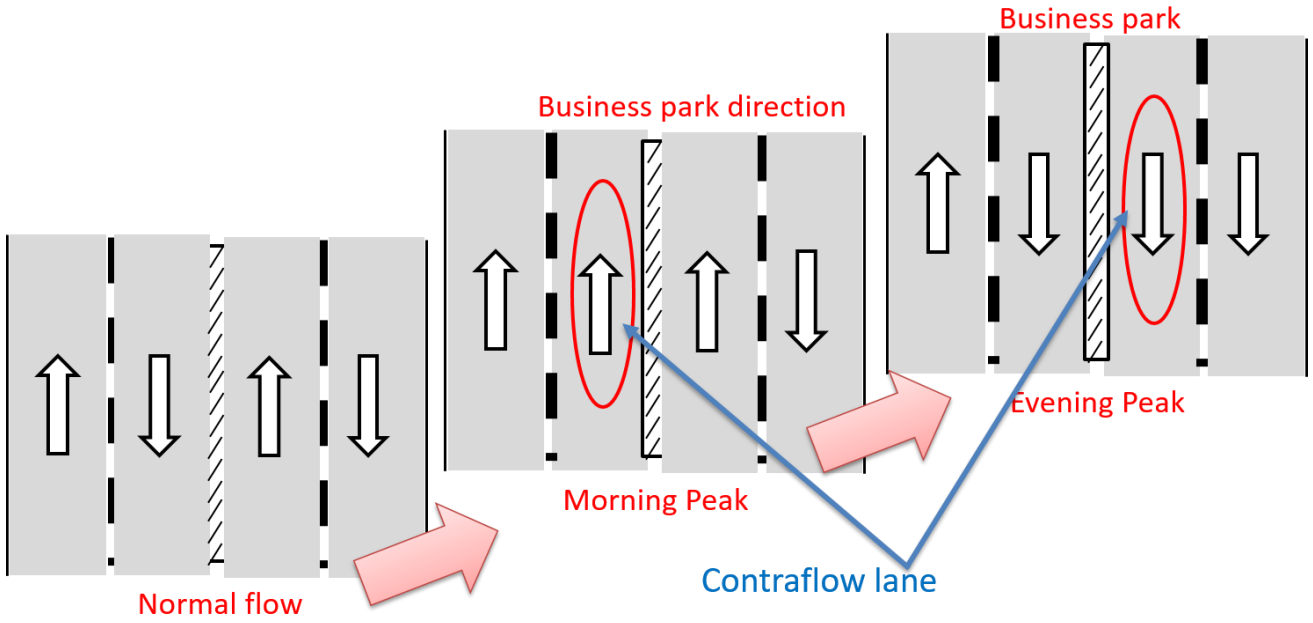
a Normal Operation



b Contraflow Operation

Source: (L. R. Kadiyali, 2017)

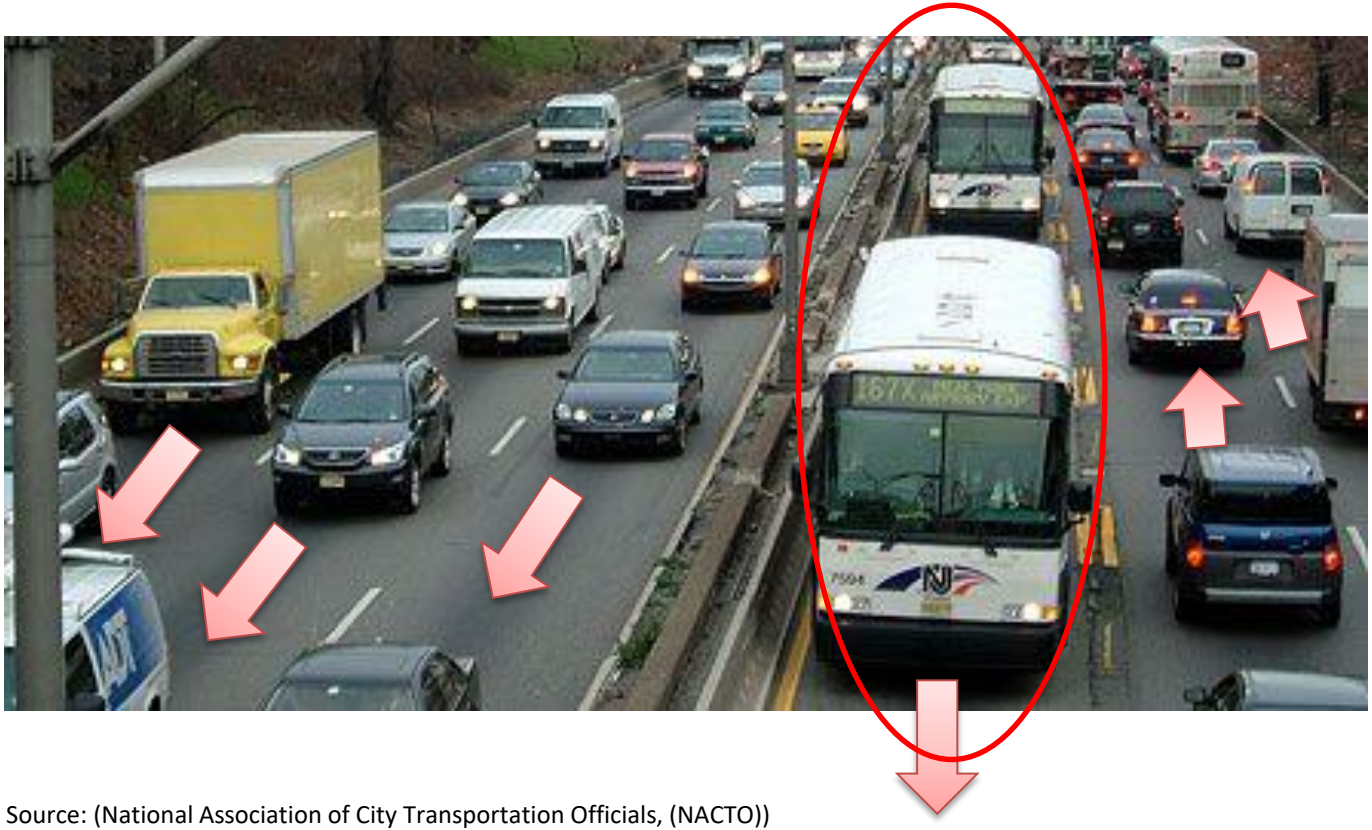
Contraflow lanes explained



During Peak hours, Contraflow lanes are usually provided for public transport vehicles, emergency vehicles or for bicyclists

Source: (L. R. Kadiyali, 2017)

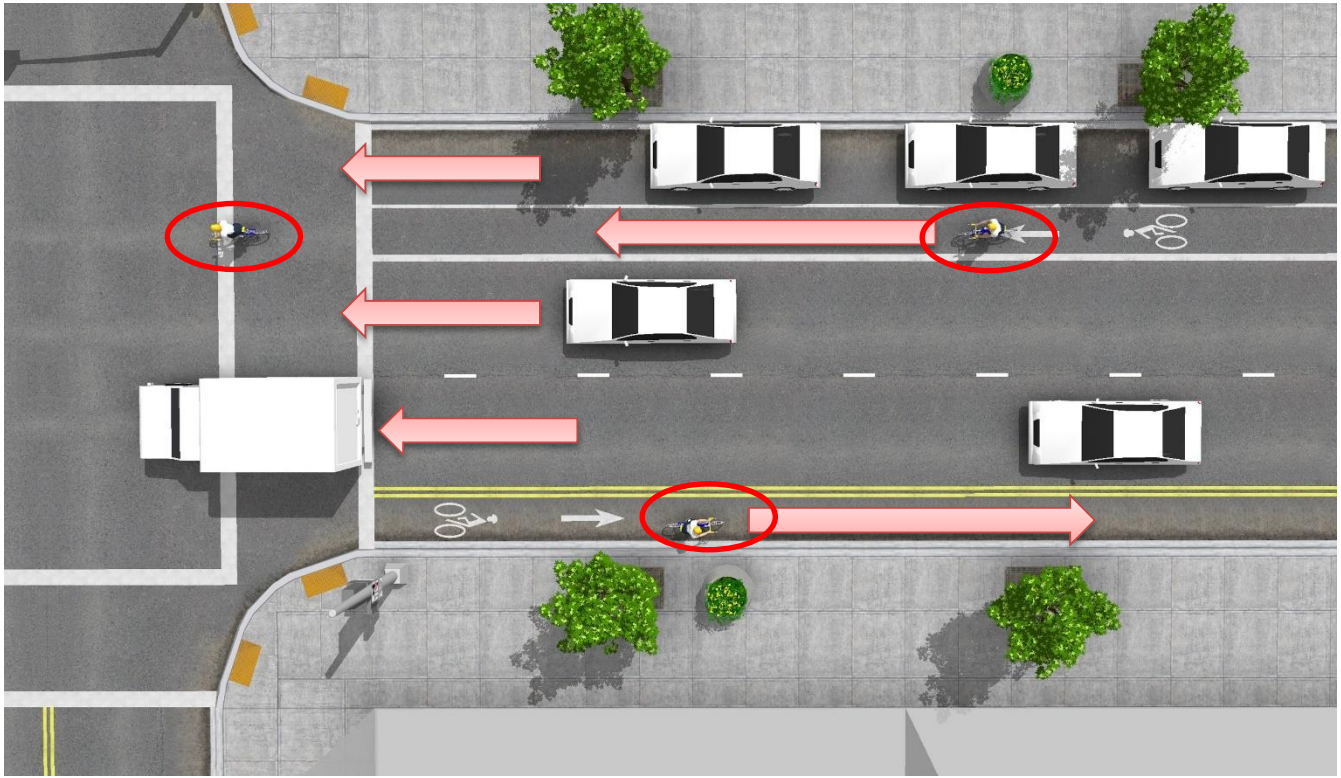
An example of a Contraflow Bus lane



An example of a Contra-flow Bus lane moving in opposite direction to the lane traffic

Source: (National Association of City Transportation Officials, (NACTO))

An example of a Contraflow Bicycle lane



An example of a Contra-flow Bicycle lane on One-way Street with Conventional Bike lanes

Source: (National Association of City Transportation Officials, (NACTO))

A Contraflow Bicycle lane in Dublin City

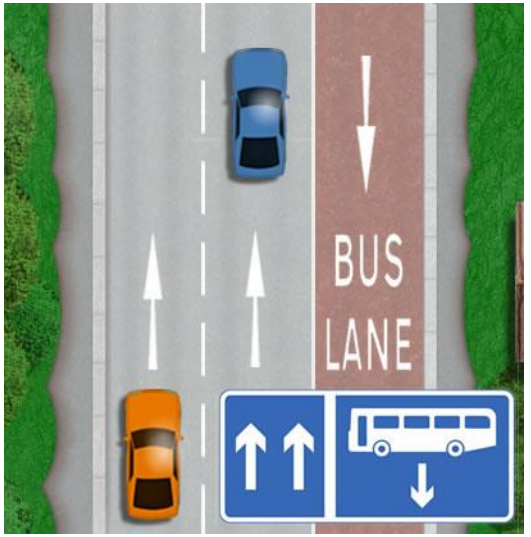


A Contra-flow Bicycle lane in Dublin City, allowing entry to the street in opposite direction only for cyclists

Source: (https://irishcycle.com/contra-flow/)

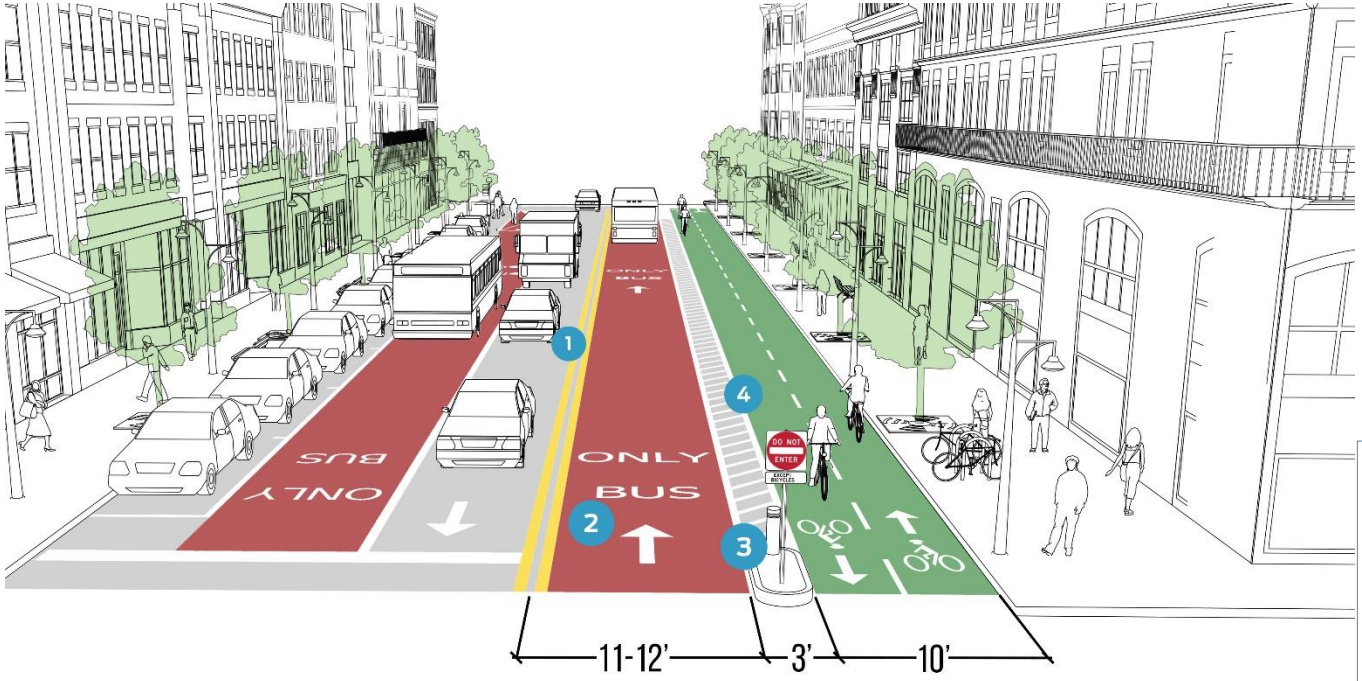
Exclusive bus lanes

- To reserve a lane of carriage way exclusively for bus traffic.
- Possible only where adequate width is available for carriageway (Minimum 3 lanes)
- Adjacent to the curb to board & embark passengers
- Exclusive bus lanes or cycle lanes are also provided as contraflow lanes



Source: (L. R. Kadiyali, 2017)

Exclusive bus lane illustration



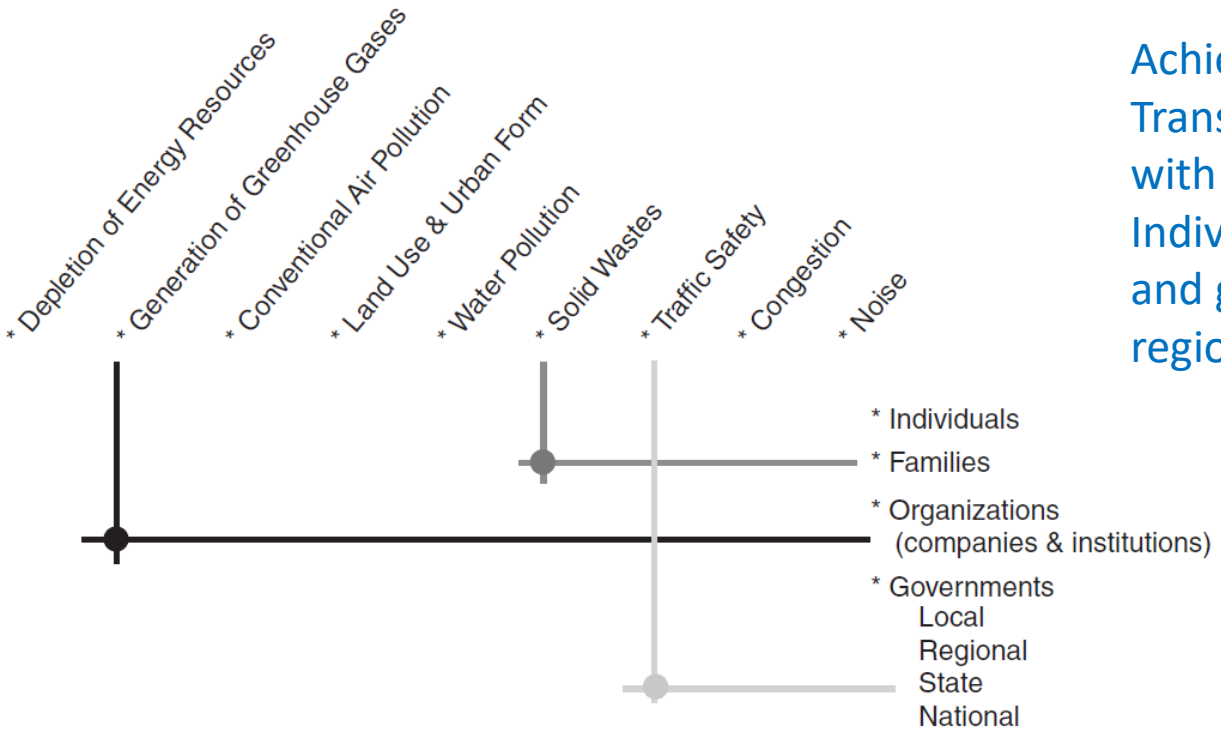
Source: (https://nacto.org/publication/transit-street-design-guide/transit-lanes-transitways/transit-lanes/contraflow-transit-lane/)

An example of Exclusive bus lane in Ahmedabad



Source: (ITDP, India)

Taming the Complexity of Sustainable Transportation



Achieving Sustainable Transportation is possible only with an integrated effort from Individuals, families, organizations and governments at the local, regional, state and national levels.

Source: (TRB, 2005)

Conclusions

- The Management strategies to Transportation systems ensures productive and cost effective solution to Peak hour congestion, delay and accidents in cities.

- Choosing the right strategy or combination of strategies could reap good results.

- Potential to reduce private vehicles on streets, congestion and delays, fuel economic benefits to passengers and low traffic emissions.

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Thank You